

TacticalBERT: Football as a Natural Language

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Abstract. The application of Large Language Models (LLMs) to sequence-based sports data offers a paradigm for football analytics, treating discrete match events as a structured language. This paper introduces TacticalBERT, a custom transformer-based architecture trained on over 4 million events from European football leagues. By integrating semantic, geometric, and contextual data into a multimodal latent representation, the model is trained via Masked Language Modeling (MLM) to capture the underlying “*grammar*” and tactical dependencies of the sport. The project explores the geometric interpretability of these embeddings, demonstrating how the model spontaneously organizes tactical taxonomies and clusters functionally similar players without explicit supervision. Furthermore, a diagnostic analysis of the multi-head attention mechanism reveals a distinct internal division of labor across layers, addressing the “*Raw Attention Problem*”. While experimental results validate the model’s capacity to decode complex match sequences (achieving a validation accuracy of 91%), the architectural limitations such as bidirectionality bias are critically discussed. Future implementations include using synchronized tracking data and advanced interpretability techniques, such as Attention Rollout and Attention Flow, to isolate the causal dependencies of scoring opportunities.

Key words: Sports Analytics, Natural Language Processing, Transformer Models, Spatio-Temporal Data, Explainable AI, Masked Language Modeling

1. Introduction

Modern football analytics has increasingly moved toward treating match events as a structured language, enabling the application of Large Language Models (LLMs) to sequence-based sports data. This project takes inspiration from the track “Cartographers of the Invisible” and investigates how these models represent tactical meaning across latent spaces. By extracting and visualizing embeddings from a transformer-based architecture, the goal is to transform abstract numerical representations into navigable semantic maps that reveal how a model structures its internal “knowledge” and its capability of understanding the game.

1.1. Literature Review

This project draws inspiration from the field of transformer interpretability, adopting a “*diagnostic*” approach to neural representations. Specifically, the project is conceptually grounded in two works:

- The geometric analysis follows the framework proposed by [4], who first demonstrated that BERT’s latent space possesses a rich, searchable geometry where semantic relationships are encoded as distances and directions.
- The investigation into attention mechanisms is motivated by the “*Raw Attention Problem*” identified by [1], which argues that simple attention weights are insufficient to track information flow due to the mixing of tokens across layers.

The landscape of football-specific language modeling is characterized by two prominent yet distinct approaches:

- **RisingBALLER** [2]: this framework adopts a player-centric paradigm where *"a player is a token and a match is a sentence"*. Its primary objective is the development of a foundational model for player data analytics. By modeling sequences of player involvements, it transforms football data analytics by learning high-level foundational features for players.
- **ScoutGPT** [3]: utilizes a generative (GPT) framework to capture player impact from team action sequences. While it also treats matches as sequences, its focus is on capture how a player's contribution adapts to a new tactical environment or different teammates.

TacticalBERT distinguishes itself from these works by focusing also on the **geometric interpretability** of tactical actions. While previous models prioritize downstream tasks like scouting or impact evaluation, this project treats the tactical event (type and context) as the primary unit of meaning. It seeks to map the "invisible" conceptual relationships—such as why certain event types cluster together or how player roles emerge spontaneously in the latent space—to ensure the model's internal logic aligns with human tactical intuition. Furthermore, the investigation goes beyond the simple analysis of weights, exploring the **Multi-Head attention** in the different layers to understand the "division of labor" internal to the model and how it encodes tactical dependencies.

2. Research Question and Methodology

The central research question asks: *How do transformer architecture structure the semantic relationships between tactical events and player roles without explicit supervision and how does the attention mechanism distribute importance across complex match sequences?* The project aims to reveal latent biases, clustering tendencies, and the underlying geometry of football knowledge, while quantifying the actual impact of events through attention flow analysis

2.1. Proposed Approach

The methodology follows a modular pipeline designed for high-dimensional data exploration:

- **Embedding Extraction:** using a pretrained BERT-style architecture, embeddings are extracted for both players and tactical events. These vectors represent the model's internal state after processing millions of event sequences.
- **Dimensionality Reduction:** techniques such as PCA and UMAP are applied to project these high-dimensional (128-D) embeddings into 2D spaces. This step is crucial for transforming mathematical representations into human-readable maps.
- **Clustering and Correlation:** K-Means algorithm is used to identify emergent groupings. Pairwise cosine similarities are computed to quantify the conceptual proximity between different football concepts.
- **Interactive Mapping:** the results are integrated into visualization maps, allowing users to explore "concept families," zoom into specific player clusters, and identify nearest neighbors in the tactical space.

- **Multi-Head Attention Analysis:** to address the limitations of raw attention weights which often become "noisy" due to residual connections and token mixing. This component retrieves specific match sequences and analyzes how the four heads across the model's layers attend to tactical dependencies, laying the groundwork for more advanced techniques like *Attention Rollout* and *Attention Flow*.

2.2. Multimodal Latent Representation

Rather than treating events as simple text tokens, the **TacticalBERT** embedding is the result of the convergence of three distinct input spaces:

- **Semantic Space:** captures the categorical event type (e.g., *Pass*, *Carry*, *Ball Receipt*).
- **Geometric Space:** encodes the discretized physical coordinates (x, y) on the pitch to provide spatial awareness.
- **Contextual Space:** incorporates situational metadata such as pressure, duration, and possession state.

The final input embedding is the **mathematical sum** of these three vectors.

3. Experimental Results

3.1. Dataset

The experimental phase utilizes a comprehensive dataset accessed from the **StatsBomb Open Data** repository¹ consisting of **1,823 matches** from the "Big Five" European leagues (Premier League, La Liga, Serie A, Bundesliga, and Ligue 1) for the **2015/2016 season**. This results in a total of more than **300k sequences** and more than **4 million individual events**.

Raw match data are parsed into a structure where a sequence represents a distinct possession phase (a sentence), and a token represents an individual event (a word). A single token is not just the event name, but a fusion of semantics, geometry and a contextual data. This structural representation is exemplified in Table 1, which illustrates the transition between two distinct possession sequences in the processed database.

Table 1: Example of Multimodal Input Sequences and Tactical Context

Sequence (Possession)	Event type (Token)	Location (x,y)	Duration (s)	Player & Context
Seq 9	Carry	50.7, 20.5	2.39	Busquets (Barca)
Seq 9	Pass	49.8, 26.5	3.22	Busquets (Barca)
Seq 9	Ball Receipt	99.8, 47.6	NaN	Ansu Fati (Barca)
– <i>Possession Change / Turnover</i> –				
Seq 10	Goal Keeper	14.8, 32.5	0.00	Courtois (Real)
Seq 10	Carry	14.8, 32.5	2.51	Courtois (Real)
Seq 10	Pass	15.7, 25.9	1.81	Courtois (Real)
Seq 10	Ball Receipt	21.7, 49.9	NaN	Varane (Real)

¹<https://github.com/statsbomb/open-data>

3.2. Model Architecture and Training Protocol

The model is a custom BERT architecture optimized for sequence modeling of match events. This choice is motivated by the model’s **encoder-only** nature, which is specifically designed for *Natural Language Understanding*. Unlike autoregressive LLMs that process sequences from left to right, BERT’s bidirectional transformer allows the model to capture the full context of a possession by looking at both preceding and succeeding events simultaneously. This is critical for football analytics, where the “*meaning*” of a tactical action (e.g., a specific pass) is often defined by the movements that occur both before and after it. The parameters of the model are visible in Table 2. The model was trained using a **Masked Language Modeling** (MLM) objective. In this setup, specific tactical events within a possession sequence are masked, and the model must predict them based on the surrounding context. This encourages the model to learn the “*grammar*” of football: the logical flow and dependencies between actions. The model parameters and training configuration are summarized in the following section.

Table 2: TacticalBERT Configuration

Parameter	Value
Model Architecture	Custom BERT (encoder-only)
Number of Layers (Depth)	4
Attention Heads	4
Embedding Dimension (H)	128
Max Sequence Length	128

3.2.1 Training Hyperparameters

The model was trained from scratch using the following protocol:

- **Optimizer:** AdamW with a learning rate of $1e - 4$.
- **Loss Function:** Weighted Cross-Entropy. Custom weights were applied to the loss to heavily penalize errors on rare, high-value events (such as goals or specific defensive interventions), ensuring the model does not ignore low-frequency but tactically significant actions.
- **Regularization:** Early Stopping was implemented with a patience of 15 epochs, monitoring the **Validation Macro F1-Score** to balance precision and recall across all event classes.
- **Batch Size:** 32.

3.3. Experimental Methodology, Metrics and Results

The experimental results demonstrate that the model successfully captures both the predictive “grammar” of football events and the latent tactical relationships between players and actions.

3.3.1 Event Prediction and Performance Metrics

To evaluate the model’s ability to learn the “language of football,” a quantitative assessment was performed based on the Masked Language Modeling (MLM) task. Metrics such as precision, recall and Macro F1-score were monitored in a hold-out validation set to measure predictive robustness across balanced and unbalanced action types. The results are visible in

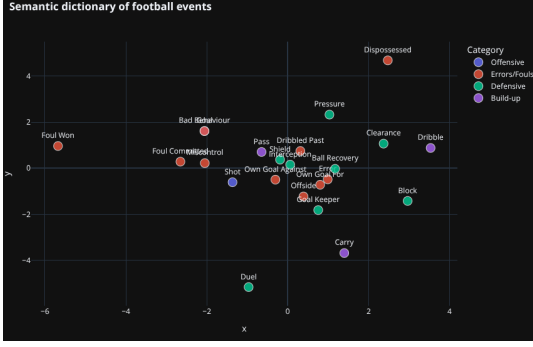
Table 3. The model reached its optimal state at epoch 72, achieving a validation accuracy of 90.55% and a Macro F1-score of 0.8049. Performance analysis across event categories indicates that the model is highly proficient at identifying frequent and structured actions, such as *Duel* (F1: 0.98), *Clearance* (F1: 0.95), and *Ball Recovery* (F1: 0.90). Rarer or more contextually ambiguous events, such as *Error* (F1: 0.45) and *Interception* (F1: 0.64), show lower performance metrics. These results suggest that the weighted cross-entropy successfully prioritized tactical sequences, though the inherent sparsity of certain events remains a limiting factor for pure predictive accuracy.

Event Type	Precision	Recall	F1-score	Support
Bad Behaviour	0.74	0.79	0.76	43
Ball Recovery	0.92	0.89	0.90	5854
Block	0.78	0.90	0.83	2070
Carry	0.81	0.97	0.89	25255
Clearance	0.94	0.97	0.95	2656
Dispossessed	0.65	0.94	0.77	1399
Dribble	0.92	0.69	0.79	1757
Dribbled Past	0.94	0.99	0.96	1102
Duel	0.98	0.99	0.98	4420
Error	0.36	0.61	0.45	23
Foul Committed	0.93	0.96	0.94	1713
Foul Won	0.96	0.99	0.98	1564
Goal Keeper	0.91	0.93	0.92	1522
Interception	0.56	0.74	0.64	1302
Miscontrol	0.59	0.88	0.70	1696
Offside	0.27	0.77	0.40	13
Own Goal For	0.80	0.67	0.73	6
Pass	0.98	0.86	0.92	52385
Pressure	0.97	0.94	0.95	17956
Shield	0.52	0.69	0.59	86
Shot	0.75	0.98	0.85	1389
Accuracy			0.91	124211
Macro Avg	0.60	0.67	0.63	124211
Weighted Avg	0.92	0.91	0.91	124211

Table 3: TacticalBERT Classification Performance (Best Epoch: 72, Val Accuracy: 0.9055, Val Macro F1: 0.8049)

3.3.2 Geometry of Tactical Events

The interpretability of the model’s vocabulary was examined through the lens of geometric projection. By applying dimensionality reduction techniques like PCA and UMAP to the static embedding layer and contextual hidden states, we measured the conceptual distance between event types to identify emergent tactical taxonomies. The analysis of geometric tactical events, Figure 1a, reveals that the model organizes the ”vocabulary” of football into a logical taxonomy. Static embeddings show clear clustering of functional families, such as offensive actions (*Shot*, *Goal*) being distinct from constructive actions. The *Pass* is positioned relatively close to the origin. This reflects its nature as a connective token, as a pass can be defensive, offensive, or neutral depending on the context. Furthermore, *Dribbled Past* is close



(a) Semantic dictionary of football events.



(b) Tactical pass clusters

Figure 1: Dimensionality reductions of events embeddings

to *Interception*. This proximity reveals that the model views these two events as one mirroring the other: both occur in the same situation where a defender attempts to interrupt the ball’s flow.

Investigation into the contextual manifolds of passes, Figure 1b, further identifies specific tactical patterns: *Tactical Resets* (backward passes for possession maintenance), *Defensive Build-up* (low-risk circulation in the defensive half), *Combinations* (short, high-pressure central play), and *Standard Progression* (forward-facing passes to enter the final third). The spatial orientation and density of these clusters, visible in Figure 2, confirm that the model’s latent space effectively maps the physical and tactical reality of the pitch.

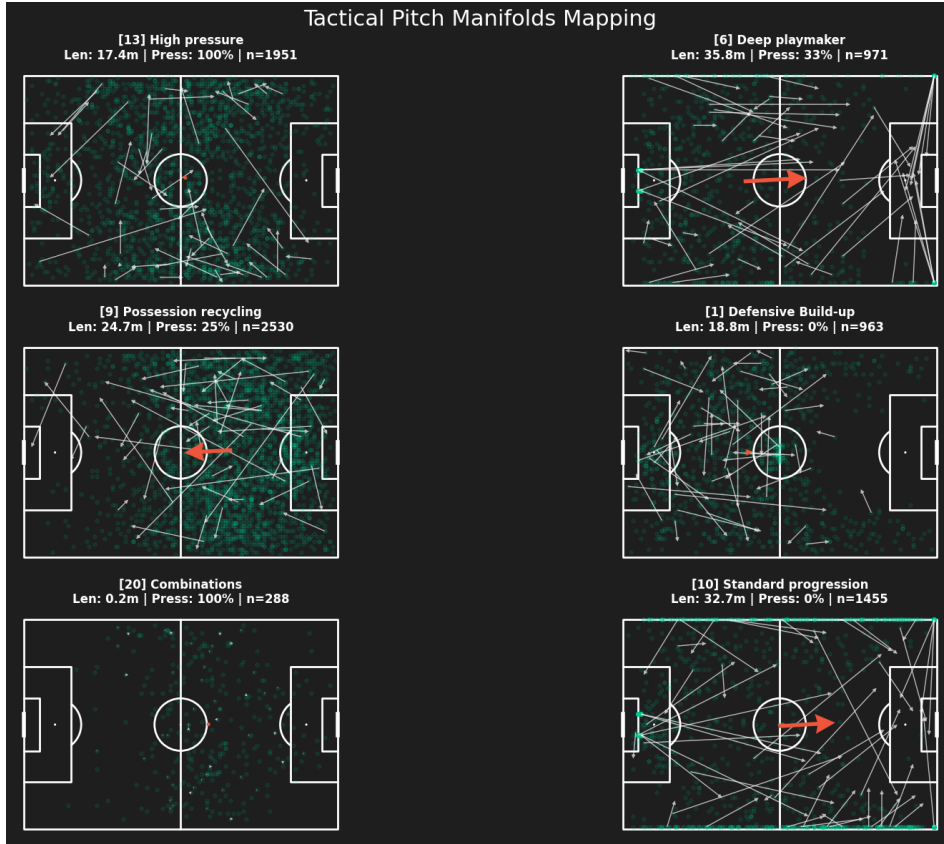


Figure 2: Pitch visualization of interesting clusters

3.3.3 Player Semantic Atlas

To assess how the model represents individual entities, a player-centric analysis pipeline was constructed. Contextual embeddings were aggregated by averaging the model’s last hidden states for each player across the entire dataset, using cosine similarity as the primary metric to quantify the proximity between different player profiles in the high-dimensional space. In the player semantic atlas, Figure 3, the high-dimensional latent space reveals the emergence of player roles without explicit supervised labels. By analyzing these contextual embeddings, the model naturally groups players with similar tactical profiles. For instance, similarity queries for a specific player profile as Messi, return a cluster of peers (e.g., Neymar, Hazard, Douglas Costa, ...) with cosine similarity scores exceeding 0.99. This indicates that the model has ”learned” player positions and functional roles solely through the observation of action sequences and their spatial contexts.

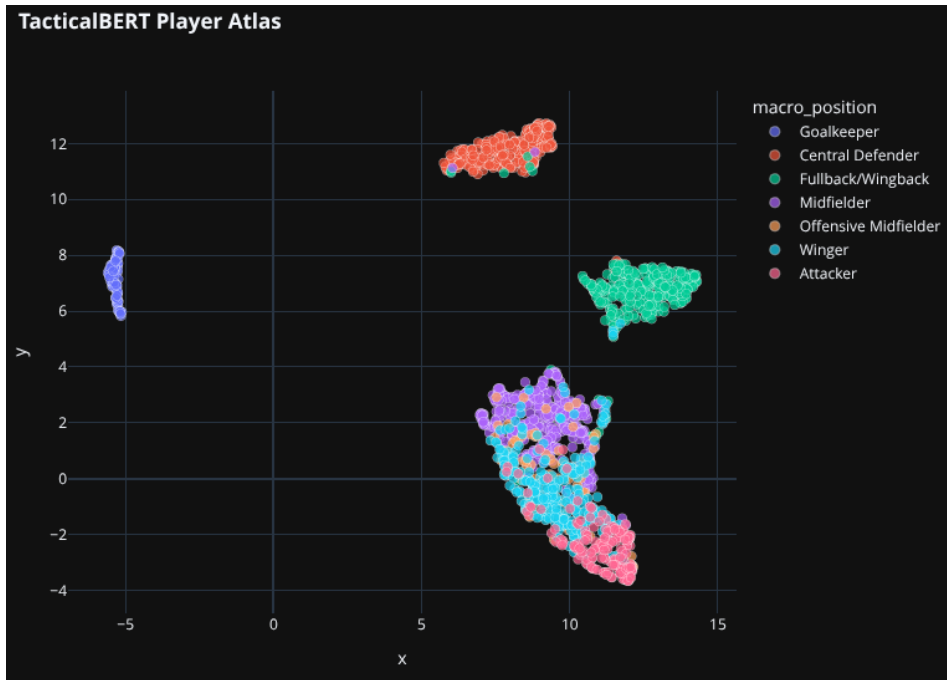


Figure 3: Player embeddings atlas

3.3.4 Multi-Head Attention Analysis

The internal logic of the transformer was examined by a diagnostic analysis of its attention heads. By extracting and visualizing the raw attention weights from all four layers during inference on specific match sequences, the distribution of focus was evaluated to determine if the model developed specialized functional behaviors or a ”division of labor.” Finally, the multi-head attention analysis identifies a specialization across the model’s architecture. By examining the behavior of the four heads across four layers, the results in Figure 4 show that individual heads develop specialized roles. To quantify this functional specialization, the analysis utilizes a suite of diagnostic metrics: **Attention Entropy** is used to distinguish between sharp, focused attention (low entropy) and broad, contextual aggregation (high entropy); **Attention Span** calculates the average step-distance between tokens to classify heads as either ”local” or ”global” in their scope. Furthermore, the model’s tactical sensitivity is measured through **Sequential focus** on immediate chronological neighbors, **Spatial alignment** between events occurring in similar pitch coordinates, and **Pressure sensitivity**, where specific

heads demonstrate an increased activation on tokens marked by defensive duress. While some heads focus on local event-to-event dependencies, others capture more global sequence relationships. This specialization addresses the "Raw Attention Problem" by demonstrating that the model does not distribute importance uniformly; instead, it utilizes its multi-head structure to partition the task of understanding complex match sequences. This internal organization confirms that the transformer architecture is actively distilling specialized tactical information at different levels of abstraction.

	ID	Layer	Head	Entropy	Span	Sequential	Spatial	Press_Sens
0	L1_H1	1	1	2.248	1.79	0.202	34.18	0.076
1	L1_H2	1	2	2.594	2.12	0.059	40.51	0.054
2	L1_H3	1	3	2.672	2.73	0.054	41.52	0.053
3	L1_H4	1	4	2.635	2.39	0.076	39.68	0.051
4	L2_H1	2	1	0.348	0.10	0.007	10.35	0.072
5	L2_H2	2	2	0.669	1.10	0.753	11.13	0.105
6	L2_H3	2	3	2.473	2.06	0.059	39.34	0.046
7	L2_H4	2	4	2.431	1.67	0.040	38.97	0.051
8	L3_H1	3	1	1.360	0.64	0.020	23.22	0.101
9	L3_H2	3	2	0.227	0.97	0.920	6.01	0.074
10	L3_H3	3	3	0.125	0.01	0.002	3.42	0.077
11	L3_H4	3	4	2.535	2.02	0.039	35.64	0.092
12	L4_H1	4	1	2.287	1.67	0.186	29.98	0.072
13	L4_H2	4	2	1.500	0.80	0.062	23.51	0.059
14	L4_H3	4	3	1.710	1.85	0.501	34.73	0.042
15	L4_H4	4	4	2.387	1.56	0.088	36.72	0.072

Figure 4: Multi-head attention analysis results

4. Conclusions

4.1. Critical Discussion

The experimental results validate the model’s capacity to internalize the complex, latent grammar of football. By mapping discrete match events into a continuous geometric space, the architecture successfully distinguishes between distinct tactical phases—such as defensive build-ups versus high-pressure combinations and accurately clusters functionally similar players within its semantic atlas.

However, the evaluation also highlights several structural and domain-specific limitations that contextualize the performance metrics. First, the lower F1-scores observed for classes such as *Error* (0.45) and *Offside* (0.40) should not necessarily be viewed as a failure of the architecture. In the highly stochastic environment of a football match, these events are inherently unpredictable anomalies. If an error were perfectly predictable from the preceding tactical sequence, players would actively avoid it. Therefore, lower accuracy on highly entropic, unpredictable events accurately reflects the physical reality and randomness of the sport.

A more significant architectural limitation stems from the bidirectional nature of the model, which is optimized via the Masked Language Modeling (MLM) task. Because the model

leverages both past and future context to reconstruct masked tokens, its predictive accuracy is sometimes inflated by "seeing the future." For example, predicting a *Shot* is heavily facilitated if the model can see a *Goal Keeper* action, a *Shield*, or an ending sequence token immediately following it. Consequently, while the current model excels at post-hoc tactical analysis and context understanding, transitioning it to a real-time predictive engine would require an autoregressive (causal) framework that strictly masks future events.

4.2. Future Works

From the results and the analysis emerge the necessity to enhance both the richness of the input data and the depth of the interpretability framework.

From a data perspective, future iterations will aim to enrich the input tokens with broader contextual information. While the current model captures the specific location of the ball event, it lacks the global spatial state of the pitch. Integrating synchronized tracking data specifically, the (x,y) coordinates of the other players and other information about the event into the token embeddings would transition the model from analyzing discrete sequential events to processing complete space-temporal frames. This would allow the model to learn the value of off-the-ball movements and structural defensive compactness.

Another interesting point to explore could be the co-occurrence bias related to squad affiliation. Since the teams have their own playing style and so do some events with a high frequency, the model might partially cluster players based on their club rather than purely on their tactical roles. To rigorously verify this, future methodologies must check for this bias by evaluating cosine similarities across different leagues and teams, analyzing players who transferred between clubs to differentiate systemic team-play patterns from individual player signatures.

In terms of Explainability, addressing the "Raw Attention Problem" remains a primary objective. The current analysis of raw attention identified the internal division of labor among heads, categorizing them into specialized roles. However, as demonstrated by Abnar & Zuidema [1], examining raw attention weights in isolation is insufficient for tracking the true impact of an early event on the final prediction. Residual connections and continuous token mixing across the transformer layers inevitably obscure the original signal. To overcome this limitation and accurately map the global propagation of information, future iterations will implement advanced interpretability techniques:

- **Attention Rollout:** Tracking how information is propagated from the input tokens through all layers by assuming a linear combination of attention weights along the computational paths.
- **Attention Flow:** Treating the transformer architecture as a flow network to compute the maximum flow of information between the input features and the final output.

By moving beyond localized head analysis to these global metrics, future research will be able to consistently identify the true causal dependencies of a possession, isolating the "Silent Architects", early actions that definitively trigger scoring opportunities.

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